



Instructor: Date:

Learner: Regional Branch:

Program: Center:

CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

What is happening to the ozone layer?



Learning Outcome

Describe environmental change in social contexts.

Ask your partner:

- How much do you know about the ozone layer?
- What effect do chemicals in the air have on the earth and the way we live?

Can Do Statements

- I can describe social and urban change, and environmental problems.
- I can talk about things that happened.
- I can talk about things that are happening and things that have happened.





Let's Explore!

Do you know what is actually happening to the ozone **layer***?

* "Layer" means a sheet or quantity of material that covers a surface.



Instructions: Read the text aloud.

Most of us have seen products that claim to be "ozone friendly" or "CFC free," but few people actually know what exactly is happening to the ozone layer. The ozone layer is a part of the earth's atmosphere (about 24 miles thick) that protects it from the sun's ultraviolet rays. Without the ozone layer, no life could survive on the planet. In the 1980s, scientists recognized that human actions had a negative effect on the global ozone layer.

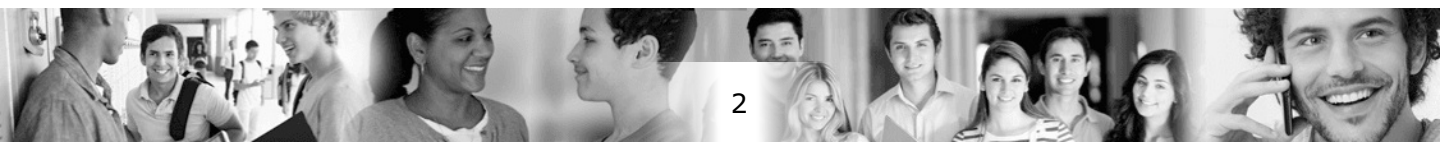
Research has shown that damage to the ozone layer has been caused by certain chemicals called CFCs (chlorofluorocarbons).

Industries have used these chemicals in refrigeration, air conditioning, cleaning materials, and aerosol sprays for a long time.

At first, scientists thought that the ozone layer was slowly thinning all over the world. However, in 1985, they discovered a large hole in the ozone that is growing above Antarctica.

Each year the ozone over this part of the earth has been reduced by over 50 percent. A thinner ozone layer means life on earth is being exposed to too much ultraviolet radiation. This can cause skin cancer, make the immune system less effective, and interfere with plants' abilities to get energy through photosynthesis. Ocean plants are also being threatened.

Many countries have tried to eliminate the manufacture and use of CFCs. However, CFCs can stay in the atmosphere for more than 100 years - a concern that will remain for a century to come.





Let's Practice!

What is the history of the ozone layer?

Instructions: Select the correct answer.

1. In the 1980s, _____ had a negative effect on the global ozone layer.
 - a. scientists recognized that human actions
 - b. scientists recognize that human actions
 - c. scientists recognizing that human actions
2. At first, _____ was slowly thinning all over the world.
 - a. scientists think that the ozone layer
 - b. scientists thought that the ozone layer
 - c. scientists have thought that the ozone layer
3. In 1985, they _____ in the ozone that is growing above Antarctica.
 - a. discovered a large hole
 - b. discover
 - c. have discovered



Let's Practice!

What are CFCs?

Instructions: Read the sentences below and circle the incorrect words.

1. Most of us have see products that claim to be "ozone friendly" or "CFC free".
2. Research has shown that damage to the ozone layer has been cause by certain chemicals called CFCs.
3. A thinner ozone layer means life on earth is being expose to too much ultraviolet radiation.
4. A lot of people have been tell about the damage to the ozone layer.
5. Ocean plants are also be threatened.





Extension Activity

Watch a movie such as *Arctic Blast*.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Watch a movie about extreme environmental catastrophes, like *Arctic Blast* or *The Day After Tomorrow*.
- Think about how that relates to what you've read about the ozone layer and the environment.
- Discuss with a partner whether you think these movies are realistic or not.

Self-Assessment

Yes

No

I watch a movie about an extreme environmental catastrophe.

☐ ☐

I reflect on what I've learned about the environment and how that relates to the movie.

☐ ☐

I discuss the movie with a partner.

☐ ☐


Target Words

- **ultraviolet** (adjective): cannot be seen by people, but responsible for making your skin darker in the sun
e.g. The ozone layer protects the earth from the sun's ultraviolet rays.
- **ray** (noun): a beam of light
e.g. The sun produces ultraviolet rays.
- **life** (noun): living things, including plants, animals
e.g. Without the ozone layer, no life could survive on the planet.
- **refrigeration** (noun): the process of keeping items cool
e.g. Industries use CFCs in refrigeration.
- **aerosol** (noun): a substance kept under pressure and able to be released as a fine spray
e.g. Industries have used these chemicals in refrigeration, air conditioning, cleaning materials, and aerosol sprays for a long time.
- **photosynthesis** (noun): the process by which a green plant uses sunlight to synthesize foods from carbon dioxide and water
e.g. This interferes with plants' abilities to get energy through photosynthesis.
- **to interfere** (verb): to prevent something from happening
e.g. This interferes with plants' abilities to get energy.
- **to eliminate** (verb): to completely get rid of something
e.g. Many countries have tried to eliminate the manufacture and use of CFCs.
- **to manufacture** (verb): to make goods by machine, usually in large numbers or amounts
e.g. They manufacture books in factories.





Learning Points

The Simple Past Tense

You use the Simple Past tense to talk about many kinds of past events: short, quickly finished actions, longer situations and repeated events.

Usually, when you use the Simple Past tense, the time of the finished action is clear from the context or explicitly mentioned in the sentence.

For example:

*In the 1980s, scientists **recognized** that human actions had a negative effect on the global ozone layer.*

Here, it is clear when the action of “recognizing” happened (in the 1980s) and that it finished in that time period.

Here is another example:

*However, in 1985, they **discovered** a large hole in the ozone that is growing above Antarctica.*

Remember that there are both regular and irregular verbs. Look at the chart for examples of the Simple Past tense verbs.

Regular Verb		Irregular Verb	
Present	Past	Present	Past
discover	discovered	think	thought
recognize	recognized	know	knew

Examples:

- *In the 1980s, scientists recognized that human actions had a negative effect on the global ozone layer.*
- *At first, scientists thought that the ozone layer was slowly thinning all over the world.*
- *However, in 1985, they discovered a large hole in the ozone that is growing above Antarctica.*





Learning Points

The Passive: The Present Continuous and Present Perfect Tenses

You use the Present Continuous to talk about an action that is happening at the time of speaking. The action or event is in progress.

*Few people actually know what exactly **is happening** to the ozone layer.*

You use the Present Perfect to talk about a finished action or event with present consequences. There is a connection between what happened in the past and the present.

*Most of us **have seen** products that claim to be "ozone friendly" or "CFC free".*

You often choose the passive form when you want to talk about an action but are not so interested in saying who or what does/did it. Passives without *agents* (the doers) are common in academic and scientific writing.

Look at the chart below for the structure of the passive voice.

Name	Structure	Example
Present Continuous	<i>am/are/is + being + past participle</i>	<i>A thinner ozone layer means life on earth is being exposed to too much ultraviolet radiation.</i>
Present Perfect	<i>have/has + been + past participle</i>	<i>Damage to the ozone layer has been caused by certain chemicals called CFCs.</i>

Examples:

- *Most of us have seen products that claim to be "ozone friendly" or "CFC free," but few people actually know what exactly is happening to the ozone layer.*
- *Research has shown that damage to the ozone layer has been caused by certain chemicals called CFCs.*
- *Industries have used these chemicals in refrigeration, air conditioning, cleaning materials, and aerosol sprays for a long time.*
- *However, in 1985, they discovered a large hole in the ozone that is growing above Antarctica.*
- *Each year the ozone over this part of the earth has been reduced by over 50 percent.*
- *A thinner ozone layer means life on earth is being exposed to too much ultraviolet radiation.*

